

Urban Heat Island (UHI) Data Challenge

An EY Learning Resource

Background

Welcome to the cutting edge of experiential learning with EY Data Challenges. We provide access to a rich repository of content from our past data challenges. This includes open-source datasets, sample code and extensive documentation, all designed to be purposefully and easily integrated into learning programs. Data challenges are more than competitions. They are a means of fostering innovation, honing problem-solving skills and encouraging collaboration. They provide a sandbox for testing theories, applying analytical skills and gaining insights that are invaluable in today's data-driven landscape. And they offer a dynamic and engaging way to complement traditional education with practical, hands-on experience.

Catering to a range of expertise levels, our past data challenges include detecting rice crops (beginner level), forecasting frog species distribution and identification of urban heat islands (intermediate level), and identifying storm-damaged buildings (advanced level). Each of these data challenges can be used in a small learning environment, such as an academic classroom or technical project team, or they can be used with a larger audience such as a university or other academic institution.

This document is meant to be used by students/participants and instructors/professors to conduct the data challenge within their learning environment. Detailed information about the content package is provided as well as scientific background on the use of satellite radar data for detection of urban heat islands. We hope that you find our EY Data Challenges to be an enjoyable and valuable learning experience!

Urban Heat Island (UHI) Challenge Objectives

The UHI challenge is aimed at participants with beginner or intermediate skills in data science. The goal of the challenge is to develop a classification model that can categorize UHI intensity into three discrete classes: Low, Moderate, and High, based on spatial and contextual geospatial data.

Participants will be given training datasets consisting of ground-truth UHI class labels with geographic coordinates (latitude and longitude) collected from two urban regions—Santiago, Chile (on January 20, 2024), and Rio de Janeiro, Brazil (on January 27, 2023). Participants will make UHI class predictions on an unseen region in Freetown, Sierra Leone, where ground-based UHI labels were collected on January 24, 2023.

Optical satellite data from the European Copernicus Sentinel-2 mission and building footprint data (3D-GloBFP) published in the Earth System Science Data (ESSD) journal will be used to create features for training the machine learning model.

Importance of Urban Heat Island (UHI) Identification

Though this topic has been widely studied and documented [References: 1,2], it is still unknown to many people and there is a need for increased awareness and accurate open-source generalized models. The primary goal of the data challenge will be to develop a digital model to predict the locations and severity of the UHI effect and to understand the drivers of this phenomenon. This micro-scale machine learning and AI model will be developed using near-surface air temperatures, building footprint data, and optical satellite data to identify vegetation and water proximity and surface state. A secondary goal of the challenge is to address the practical application of the output models for local decision-makers, scaling such solutions to other cities around the world, and considerations for improved model accuracy.

The UHI effect is typically modeled using coarse satellite-based measurements of surface temperatures such as those produced by NASA's Landsat mission. Such models do not reflect the near-surface micro-climate air temperature that most impacts the human population. Therefore, there is a need for simplified, yet accurate, UHI models to allow urban planners and city managers to better understand the location and severity of UHI issues in their city and the drivers of urban heating. Such models will bring attention to this important global sustainability problem and may force decisions to alter existing urban plans or influence future ones. In addition, such models can be used to understand the impact of natural areas and support their preservation and future expansion.

Dataset Summary

Temperature Data

Near-surface temperature data was collected by CAPA Strategies using a ground traverse (Figure 1) with vehicles and bicycles on a single day in Santiago, Chile, on January 20, 2024 (Figure 2), and in Rio de Janeiro, Brazil, on January 27, 2023 (Figure 3). This data collection effort resulted in 21,662 data points for the Chile location and 28,488 data points for the Brazil location, which will be the focus of this data challenge.



Figure 1. Ground-level temperature data was collected by CAPA Strategies and community volunteers using temperature recording devices mounted to cars and bikes. This data collection campaign was part of the international “Heat Watch” program. Credit: CAPA Strategies, LLC.



Figure 2. Data was collected across Santiago, Chile, on January 20, 2024, between 3:00 pm and 4:00 pm. The data (21,662 points) was converted to a UHI class for the purpose of this data challenge. The image above shows areas of lower temperature values (cool spots) in yellow and areas of higher temperature values (hot spots) in dark red.



Figure 3. Data was collected across Rio, Brazil, on January 27, 2023, between 3:00 pm and 4:00 pm. The data (28,488 points) was converted to a UHI class for the purpose of this data challenge. The image above shows areas of lower temperature values (cool spots) in yellow and areas of higher temperature values (hot spots) in dark red.

For this challenge, we have created a unique UHI index for every data point location. This index reflects the local temperature at the data point location compared to the city's average temperature across all data points during the time window of the data collection. Based on this UHI Index, the data was separated into three UHI classes: Low, Medium and High. Though this is not a perfect approach to modelling the complex urban heating dynamics of a city, it will provide a reasonably accurate model of urban heat islands in the city at the time of day consistent with the data collection. In an ideal situation, time series data would be collected at thousands of locations across the city and weather data (e.g., wind speed, wind direction, solar flux) would be added to the model to yield more accuracy and allow for consideration of natural variability.

UHI Index = (Temperature at a given location) / (Mean temperature for all locations)

UHI Class: Low (UHI Index ≤ 0.98), Medium ($0.98 < \text{UHI Index} < 1.02$), High (UHI Index ≥ 1.02)

The chosen UHI index classification serves as a crucial metric for assessing the intensity of heat within different urban zones of the city. For comparison, most literature calculates a UHI index based on temperature differences between inner city locations and rural locations far outside of the city. Since we did not have data from rural locations, we created a unique UHI index classification that reflects the variability of temperatures within our collected dataset. As an example, a UHI index value of 1.0 (Medium UHI Class) suggests the local temperature is the same as the mean temperature of all collected data points. UHI index values above 1.02 (High UHI Class) are consistent with hotspots above mean temperature values and UHI index values below 0.98 (Low UHI Class) are consistent with cooler locations in the city. Participants will use their models to predict these UHI values and their associated classifications (Low, Medium, High) across the city. Figures 4 and 5 show histograms of UHI values for the ground-based training data collected in Chile and Brazil.

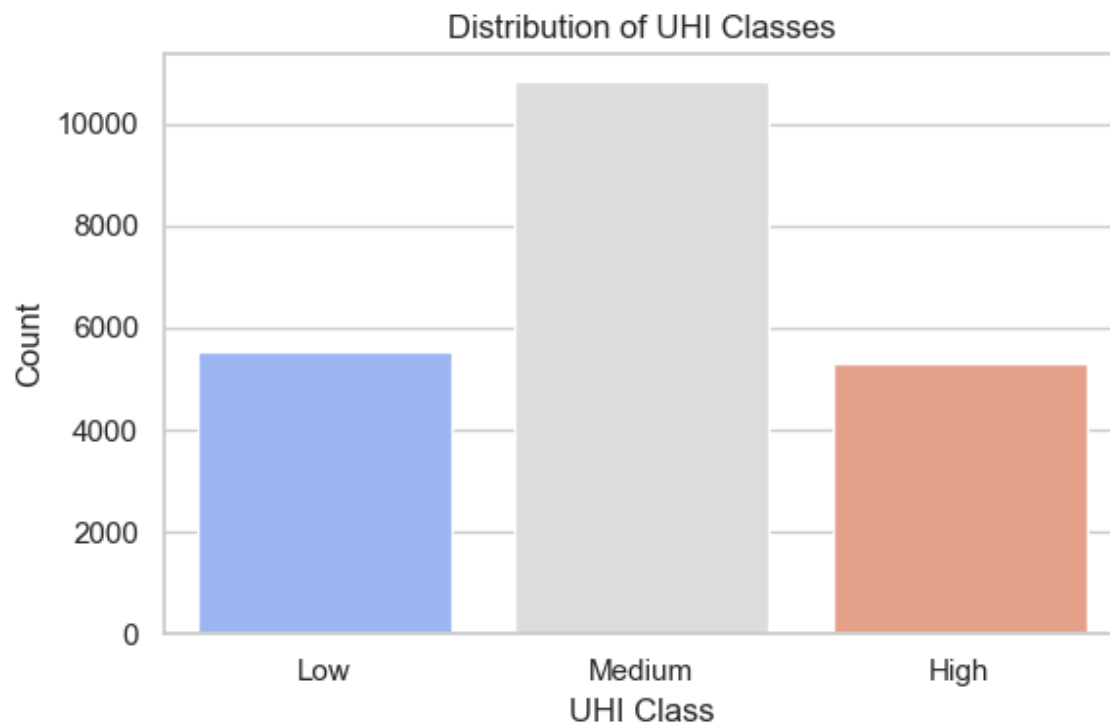


Figure 4. Histogram of UHI Classes based on 21,662 data points collected in Santiago, Chile, on January 20, 2024. The chart shows a higher frequency of Medium UHI values, while Low and High UHI values are nearly equal, indicating a mix of heat exposure across the region. Although this histogram is based on categorical UHI classes rather than exact temperatures or index values, it still reveals meaningful spatial variation in urban heat intensity.

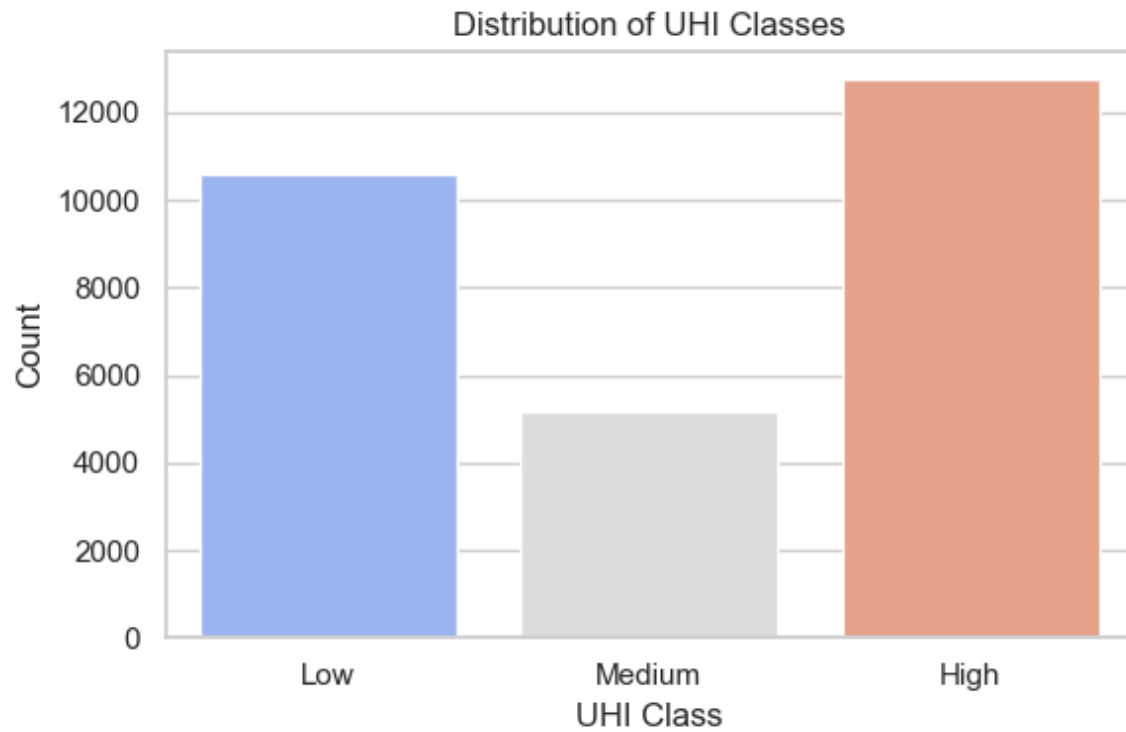


Figure 5. Histogram of UHI Classes based on 28,488 data points collected in Rio de Janeiro, Brazil, on January 27, 2023. The distribution clearly shows a concentration of data points in the High and Low UHI classes, with significantly fewer points in the Medium category. This bimodal pattern reflects the sharp contrasts in land use across the region - dense urban developments contributing to higher UHI values, and large green or coastal areas contributing to lower values. The pronounced variation demonstrates the strong influence of built environment and natural features on localized urban heating patterns.

Building Footprints

Many studies suggest that the density and height of buildings in a city influence ground temperatures and ultimately contribute to the UHI issue. [Reference 3] This effect is typically driven by buildings blocking the flow of air and adding waste heat. For this challenge, we have provided a building footprint and height dataset (Figure 6) [Reference 4]. Such information could be used in your digital model as a feature that drives local urban heating.



Figure 6. Higher building density is known to increase local air temperatures and contribute to the UHI effect. Data challenge participants will be given building footprint data for consideration in their digital models. The above footprints correspond to the Rio, Brazil region.

Satellite Data

Satellite data can be quite beneficial for understanding the location and severity of urban heat islands. Missions such as NASA's Landsat and European Space Agency's (ESA) Sentinel-2 provide continuous global coverage at regular revisit rates and open and free access to the datasets on cloud computing frameworks. These datasets provide a unique look at urban areas and often measure regions not covered by in-situ measurements. But these datasets also have limitations that must be considered when using them for UHI models. For example, these optical satellites cannot penetrate clouds or vegetative cover, so this often leads to data gaps and decreased data quality. In addition, these satellites have consistent acquisition times near the middle of the day which does not usually coincide with collected ground data or the times of maximum urban heating. So, some of these factors should be considered when using satellite data in UHI models.

The launch of the European Copernicus Sentinel-2 missions in 2015 and 2017 provides optical data at 10-meter spatial resolution and a revisit every 10 days with one mission and every 5 days with two missions. This free and open data is readily available from the Microsoft Planetary Computer (<https://planetarycomputer.microsoft.com/catalog>). But optical data cannot penetrate clouds, so it is necessary to filter out clouds or select scenes that have very low levels of cloud cover. For this challenge, we have provided sample Sentinel-2 Python notebooks that demonstrate cloud filtering, the creation of cloud-filtered median mosaics, the creation of spectral index products, and the creation of GeoTIFF output products for further analyses.

These satellite-based products can be used to assess the impacts of vegetation extent, water, or urban density on urban heating. It is well known [References: 5,6] that proximity to vegetation (green space), proximity to water, and local urban density contribute to the effects of urban heating. Below (Figures 7 and 8) are examples of Sentinel-2 mosaic products showing the spatial variation of the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Buildup Index (NDBI) over one of the data collection regions. This information can be used in a digital UHI model as vegetation, urban density, and water can impact local urban heating.

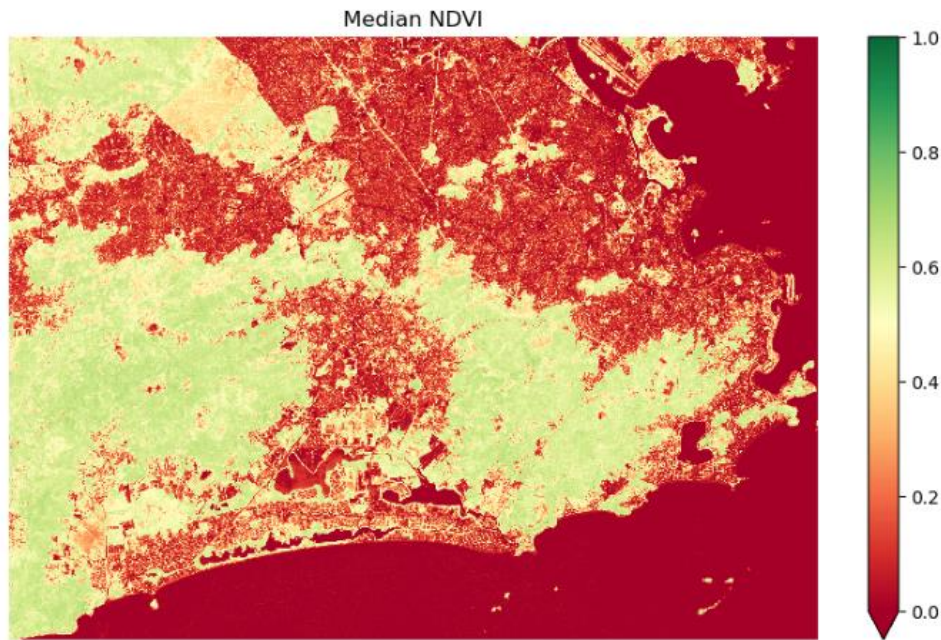


Figure 7. Sentinel-2 Normalized Difference Vegetation Index (NDVI) over the data collection region in Brazil. This cloud-filtered product is based on a median mosaic of 9 images from December 1, 2022 to April 1, 2023. Areas of light green are consistent with the presence of vegetation. Areas of dark red are consistent with dense urban environments or water.

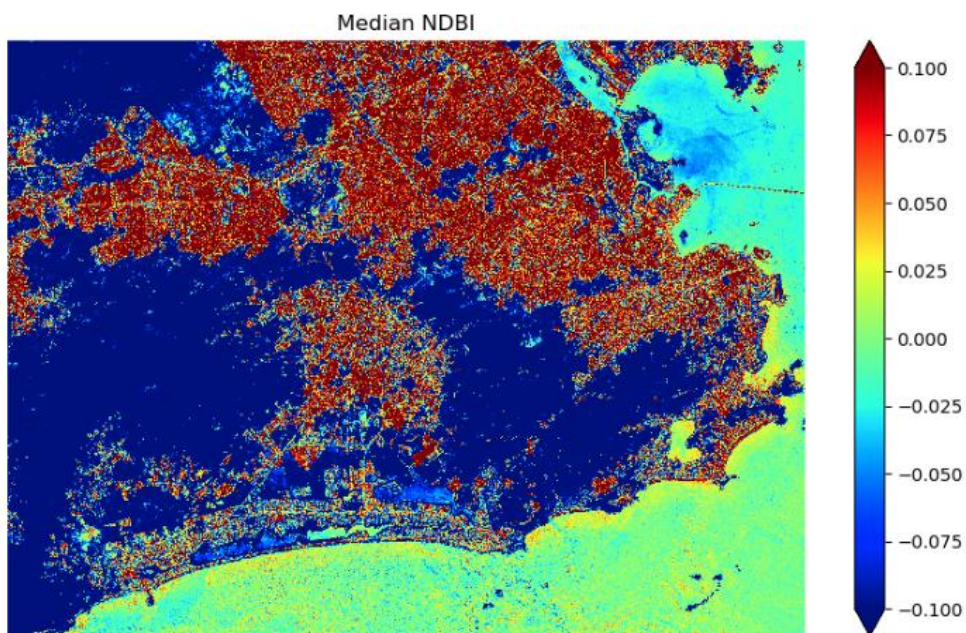


Figure 8. Sentinel-2 Normalized Difference Buildup Index (NDBI) over the data collection region in Brazil. This cloud-filtered product is based on a median mosaic of 9 images from December 1, 2022 to April 1, 2023. Areas of dark blue are consistent with the presence of vegetation. Areas of dark red are consistent with dense urban environments. Areas of green or light blue are consistent with water.

References

- [1] Poumadère M, Mays C, Le Mer S, Blong R. **The 2003 Heat Wave in France: Dangerous Climate Change Here and Now.** *Risk Analysis*, Vol. 25, Issue 6, Dec 2005, pp. 1483-1494. <https://doi.org/10.1111/j.1539-6924.2005.00694.x>
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- [3] Lee S, Kim D. **Multidisciplinary Understanding of the Urban Heating Problem and Mitigation: A Conceptual Framework for Urban Planning.** *Int J Environ Res Public Health*. 2022 Aug 18;19(16):10249. <https://doi.org/10.3390/ijerph191610249>
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- [6] Voelkel, J., & Shandas, V. (2017). **Towards Systematic Prediction of Urban Heat Islands: Grounding Measurements, Assessing Modeling Techniques.** *Climate*, 5(2), 41. <https://doi.org/10.3390/cli5020041>

Package Content

1. Challenge Overview (**UHI_Challenge_Overview.docx**) – this document
2. Benchmark Notebook (**Sample_Model_Notebook.ipynb**) – a Python notebook that provides a starting point for students and participants
3. Sentinel-2 satellite data sample notebook (**Sample_GeoTiff_Creation.ipynb**) - a Python notebook that outputs GeoTIFF files for 3 locations (Chile, Brazil, Sierra Leone) with 3 spectral indices for a specific date. These output files are used by the Benchmark Notebook. NOTE: These output files are rather large (100MB to 500MB each).
4. Sentinel-2 satellite data sample notebook (**Sample_Median_Mosaic.ipynb**) - a Python notebook that demonstrates cloud filtering, generation of a median mosaic, and outputs a GeoTIFF file with 12 spectral bands and 3 spectral indices. This output can be used by students and participants to expand their model and explore more satellite data.
5. Requirements (**requirements.txt**) – a text file listing Python package dependencies for the benchmark notebook execution

Data Folder:

1. Training datasets:
 - a. **“Sample_Brazil_uhi_data.csv”** - CSV file that contains locations (latitude and longitude) and the corresponding UHI class for a region in Brazil
 - b. **“sample_chile_uhi_data.csv”** - CSV file that contains locations (latitude and longitude) and the corresponding UHI class for a region in Chile
2. Building Footprint datasets:
 - a. **“Brazil Building Footprints”** - a folder with building footprint shapefiles (3 total) for the Brazil region
 - b. **“Chile Building Footprints”** - a folder with building footprint shapefiles (3 total) for the Chile region
 - c. **“Sierra Leone Building Footprints”** - a folder with building footprint shapefiles (3 total) for the Sierra Leone region
3. Validation dataset: **“Validation_Dataset.csv”** - CSV file that contains 14105 locations (latitude and longitude) in Sierra Leone for making prediction of UHI class

Instructor Folder:

- a) **“UHI Class Prediction Evaluation Notebook.ipynb”** – an automated Python notebook evaluation script for instructors/professors. The notebook produces an F1-score along with details about precision and recall per UHI class.
- b) **“Ground_Truth.csv”** – a CSV file containing known UHI classed of all the locations of the validation dataset. This file is used by the Evaluation notebook (above).

Instructions for Students / Participants

a) Understand the problem statement and dataset

The objective of this data challenge is to build a machine learning model to predict Urban Heat Island (UHI) intensity levels using three discrete classes: Low, Medium, and High, based on spatial and contextual geospatial data. This exercise will use ground-truth data of UHI class labels collected from two urban regions—Santiago, Chile and Rio de Janeiro, Brazil. Please refer to the training datasets which consist of locations (latitude and longitude) with a tag specifying its UHI intensity class.

b) Understand the feature dataset

For building the machine learning model, we need to extract the features from the Sentinel-2 satellite dataset and building footprint dataset. Let’s understand them one by one.

Sentinel-2 Dataset:

There are two common methods to extract band values from a Sentinel-2 satellite dataset. These are using an API or using a pre-processed GeoTIFF image. Students may choose any of these approaches based on their convenience. Given the large size of our dataset, using the API method can be time-consuming and computationally intensive. Therefore, we have provided a sample GeoTIFF notebook that outputs images of the training regions (Chile, Brazil) and validation region (Sierra Leone).

Building footprint dataset:

This dataset can be leveraged for creating the features like building density, building height, etc. This dataset has been provided for the training regions (Chile, Brazil) and validation region (Sierra Leone) in shapefile format (.shp). Please refer to data folder and the footprint sub-folders.

c) Set up the development environment

Building a machine learning model requires a Python notebook development environment (e.g., Jupyter Lab) and a Python computing environment with the appropriate Python libraries. Please refer to the file “*requirements.txt*”, which contains a list of necessary Python libraries required to run the benchmark notebook. You can use a local computing environment (e.g., PC, Mac) or any cloud environment (e.g., Azure Cloud, Google Cloud). For reference, the sample notebooks have been successfully tested on a local laptop computer with 4 cores and 32GB memory, so a paid cloud environment is not required.

d) Build the model

Once your development environment is ready, you need to create a Python notebook and start building the model. Please refer to the file “*Sample_Model_Notebook.ipynb*”, where we have demonstrated a basic machine learning workflow, designed to help students gain practical experience with real-world datasets. The focus of this example is to predict UHI class using features from the Sentinel-2 satellite dataset and building footprint dataset as predictor variables. This notebook is simplified and intended as a learning tool for students to experiment with remote sensing data and machine learning techniques. This notebook should be considered as a starting point only and students should extend this model to build a more robust model by running multiple experiments.

e) Make the prediction on the validation dataset

After building the model, students need to validate the model performance on the dataset of a different location. To validate the model, students need to use their model to make predictions about the UHI class of test coordinates we have provided in the “*Validation_Dataset.csv*” file and share the predicted results to the instructor/professor.

Instructions for Instructors / Professors

a) Understand the ground truth dataset

Please be familiar with the ground truth data of the locations in the validation dataset. You can refer to the file “*Ground_Truth.csv*” file which consists of geolocations and their corresponding UHI class (Low/Medium/High).

b) Evaluate the predicted result

After the submission of predicted results by students in a csv file “*Predicted_Dataset.csv*”, you can assess their performance by executing the Python notebook “*UHI Class Prediction Evaluation Notebook.ipynb*” to generate the model evaluation metrics including: Accuracy, Precision, Recall, and F1 Score. Out of various metrics, the F1 Score is considered as the most appropriate metric to compare the model performance.